DS231: Data Visualization (Tableau/Power-BI/Excel)

Rational

This course will help learners to:

1. Understand the importance of data visualization in data analysis and decision-making.
2. Effectively use Tableau and Power BI for data preparation, exploration, and visualization.
3. Create a variety of data visualizations, including charts, graphs, dashboards, and interactive reports.
4. Apply best practices in data visualization design to convey insights clearly and accurately.
5. Interpret and communicate findings from visualized data to support decision-making.
6. Work with real-world datasets to practice data visualization techniques.
7. Identify and troubleshoot common issues in data visualization projects.
8. Use data storytelling techniques to engage and inform diverse audiences.
9. Incorporate advanced interactivity and user parameters in Tableau and Power BI.
10. Demonstrate proficiency in connecting to and transforming data from various sources.
11. Apply principles of data ethics and privacy in data visualization projects.
12. Collaborate on group data visualization projects and provide constructive feedback.
13. Evaluate and select appropriate visualizations based on data characteristics and objectives.
14. Understand the limitations and constraints of data visualization tools and techniques.
15. Incorporate custom scripting and calculations in Tableau and Power BI.
16. Create responsive and mobile-friendly data visualizations for diverse platforms and devices..

E1.11.2 Learning outcomes

By the end of the course-tenure the students should be able to:

• Understand the importance of data visualization in data analysis and decision-making.

• Effectively use Tableau and Power BI for data preparation, exploration, and visualization.
• Create a variety of data visualizations, including charts, graphs, dashboards, and interactive reports.
• Apply best practices in data visualization design to convey insights clearly and accurately.
• Interpret and communicate findings from visualized data to support decision-making.
• Work with real-world datasets to practice data visualization techniques.
• Identify and troubleshoot common issues in data visualization projects.

Course content

Unit I: Fundamentals of Data Visualization with Tableau and Power BI

- Introduction to the core principles of data visualization.
- Understanding the relationship between data analytics and data visualization.
- Installation of Tableau and Power BI software.
- Exploration and navigation of the Tableau and Power BI workspaces.
- Collecting and assembling data from multiple sources.
- Importing and preparing data within the Tableau and Power BI environments.
- Overview of different file types used in Tableau and Power BI projects.

Unit II: Essential Principles of Visualization Creation

- Creating effective visualizations to represent data.
- Principles of data filtering.
- Building data hierarchical structures for deeper insights.
- Creating sets and handling date data.
- Selection and construction of various chart and graph types.
- Examination of effective and ineffective visualization examples.
- Emphasis on design principles for impactful data visualizations.

Unit III: Visual Analytics with Tableau and Power BI

- Principles for simplifying and decluttering data visualizations.
- Customizing data using different functions (number, string, logical, date, aggregate).
- Introduction to statistics and statistical tools.
- Basic forecasting techniques.

Unit IV: R Integration with Tableau and Power BI

- Steps for integrating R with Tableau and Power BI.
- Exploring linear regression.
- Performing "Market Basket Analysis" in Tableau.
- Introduction to K-means Clustering.

Unit V: Dashboard Creation and Storytelling with Tableau and Power BI

- Creating interactive dashboards.
- Adding, removing, and organizing views and worksheets.
- Utilizing dashboard actions for user interactivity.
- Introduction to storytelling with data.
- Creating compelling data stories to engage and inform the audience.

Unit VI: Best Practices in Data Visualization

- Understanding the art and science of data visualization.
- Applying principles for effective data storytelling.
- Designing for diverse audiences and personas.
- Collaborative group projects and evaluation.
- Recognizing the limitations and constraints of Tableau, Power BI, and data visualization tools.
- Real-world case studies and contemporary trends in data visualization.

Unit VII: Real-World Data Visualization Projects

- Hands-on projects with real-world datasets.
- Troubleshooting and problem-solving in data visualization projects.
- Engaging in group project work and delivering effective presentations.
- Providing constructive feedback and peer reviews.
- Demonstrating proficiency in data visualization techniques.

Unit VIII: Data Visualization for Informed Decision-Making

- Constructing interactive dashboards for business intelligence.
- Implementing user interactivity and parameterization in Tableau and Power BI.
- Leveraging data insights for data-driven decision-making.
- Ethical considerations within data visualization.
- Concluding with final group project presentations.
- Course review, feedback collection, and planning for future steps.

E1.11.4 Teaching and Learning Methods

Lectures:45Hrs Tutorials:30Hrs, Labs:30Hrs, Assessments:30Hrs, NATIONAL HRS: 135

E1.11.4 Assessment

- Continuous Assessment 40%
  - 20% for hands-on assessments
  - 20% for non-hands-on assessments
- Final Exams 60%
  - 60% for theory assessment or
  - 40% theory assessment and 20% practical assessment

E1.11.5 Recommended Readings

- Ryan Sleeper (2018), Practical Tableau: 100 Tips, Tutorials, and Strategies from a Tableau Zen Master, O'Reilly Media, Inc., Sebastopol.
- Seema Acharya and Subhashini Chellappan (2017), Pro Tableau: A Step-by-Step Guide, Apress
- Media, LLC, DOI 10.1007/978-1-4842-2352-9
- Murray, D. (2021). Tableau Your Data! (3rd ed.). Wiley.
- Sleeper, R. (2020). Practical Tableau: 100 Tips, Tutorials, and Strategies from a Tableau Zen Master. O'Reilly Media.
- Puls, K., & Escobar, M. (2016). M is for (Data) Monkey: A Guide to the M Language in Excel Power Query. Holy Macro! Books.
- Seamark, P., & Ferrari, A. (2019). Power BI DAX: Data Analysis Expressions. Holy Macro! Books.